

IT3681 MOBILE APPLICATIONS DEVELOPMENT LABORATORY

LIST OF EXPERIMENTS:

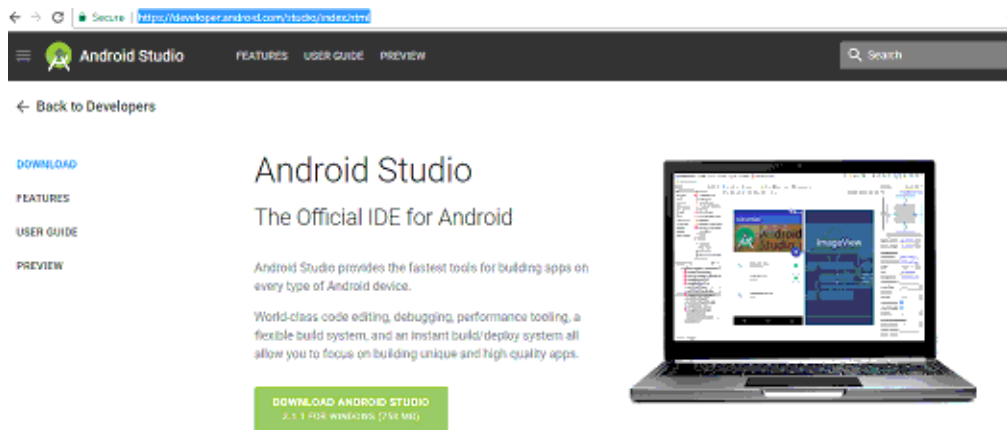
- ☐ Study and installation of Flutter/Kotlin multi-platform environment
- ☐ Develop an application that uses Widgets, GUI components, Fonts, and Colors.
- ☐ Develop a native calculator application.
- ☐ Develop a gaming application that uses 2-D animations and gestures.
- ☐ Develop a movie rating application (similar to IMDB)
- ☐ Develop an application to connect to a web service and to retrieve data with HTTP.
- ☐ Develop a simple shopping application.
- ☐ Design a web server supporting push notifications.
- ☐ Develop an application by integrating Google maps
- ☐ Mini Projects involving Flutter/Kotlin multi-platform

EX. NO: 1 STUDY AND INSTALLATION OF FLUTTER/KOTLIN MULTI-PLATFORM ENVIRONMENT

PROCEDURE:

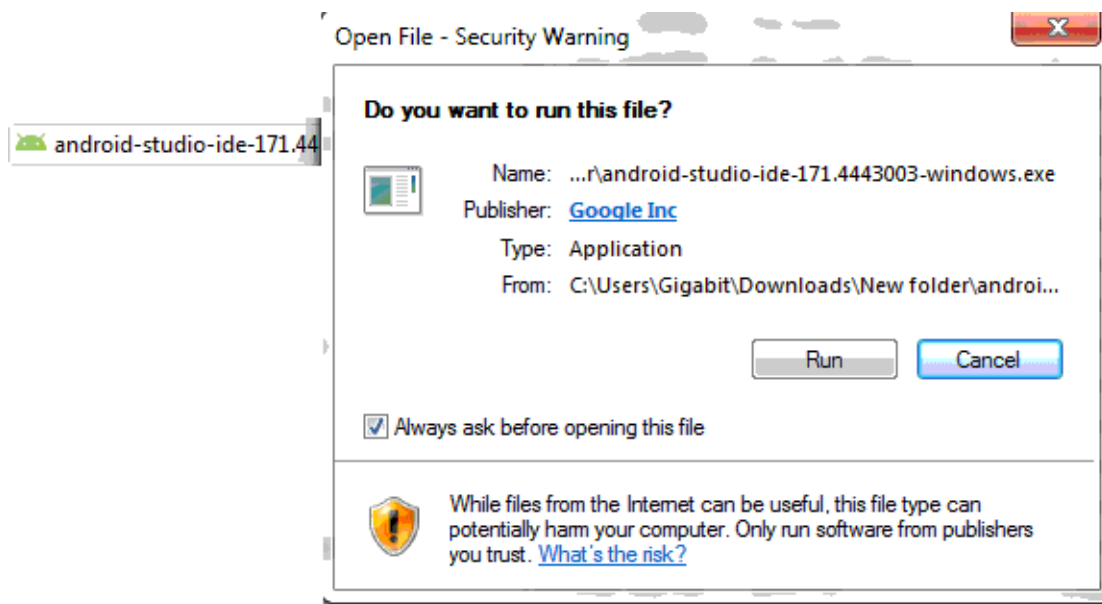
(i) To develop an android application using Kotlin, we need a Android application development IDE such as Android Studio.

(ii) Android Studio is an IDE that allows us to develop application at single platform. We can download Android Studio set from Android Studio official link [Download Android Studio](https://developer.android.com/studio/index.html).

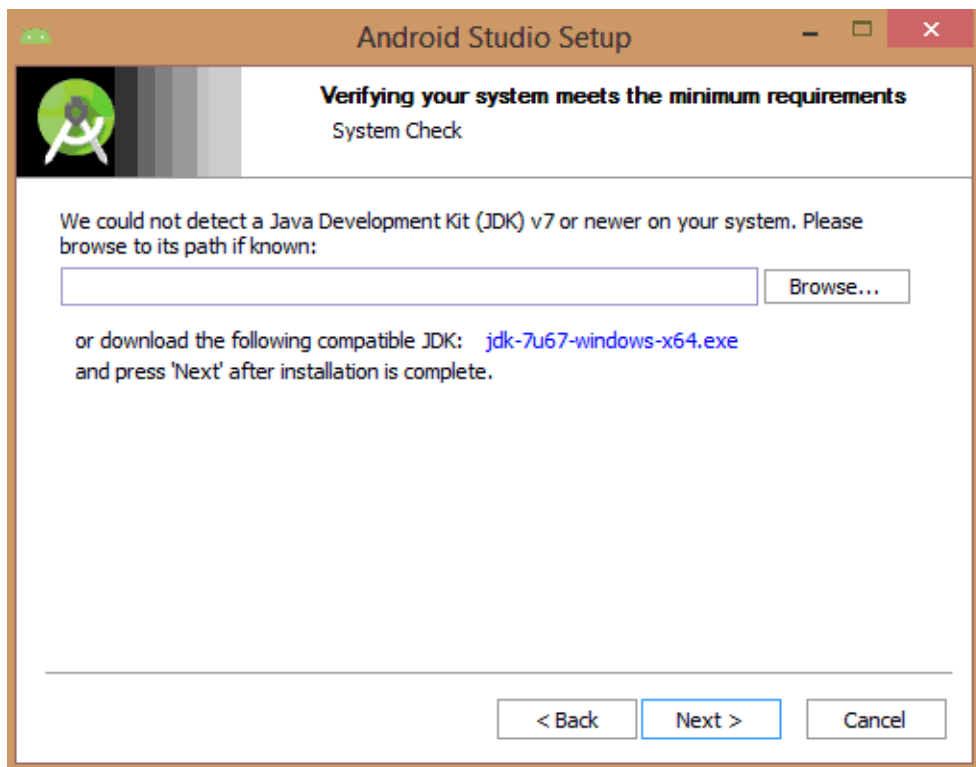
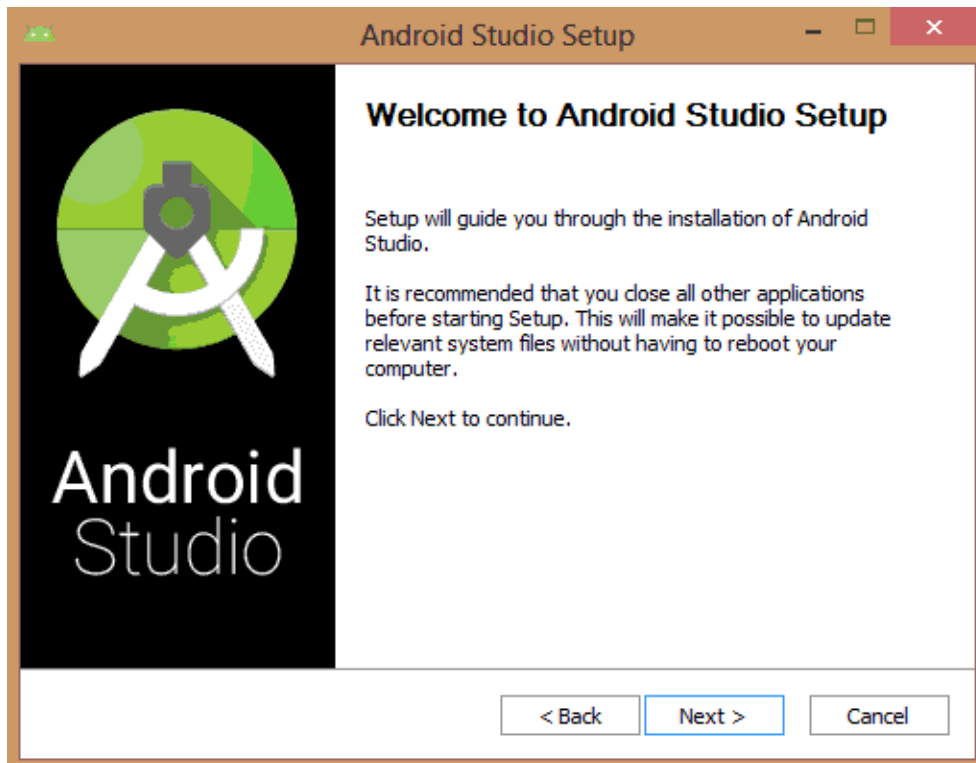


Before installing Android Studio on our machine, we must have Java installed. Look a reference here <https://www.javatpoint.com/how-to-set-path-in-java> to set path of JDK.

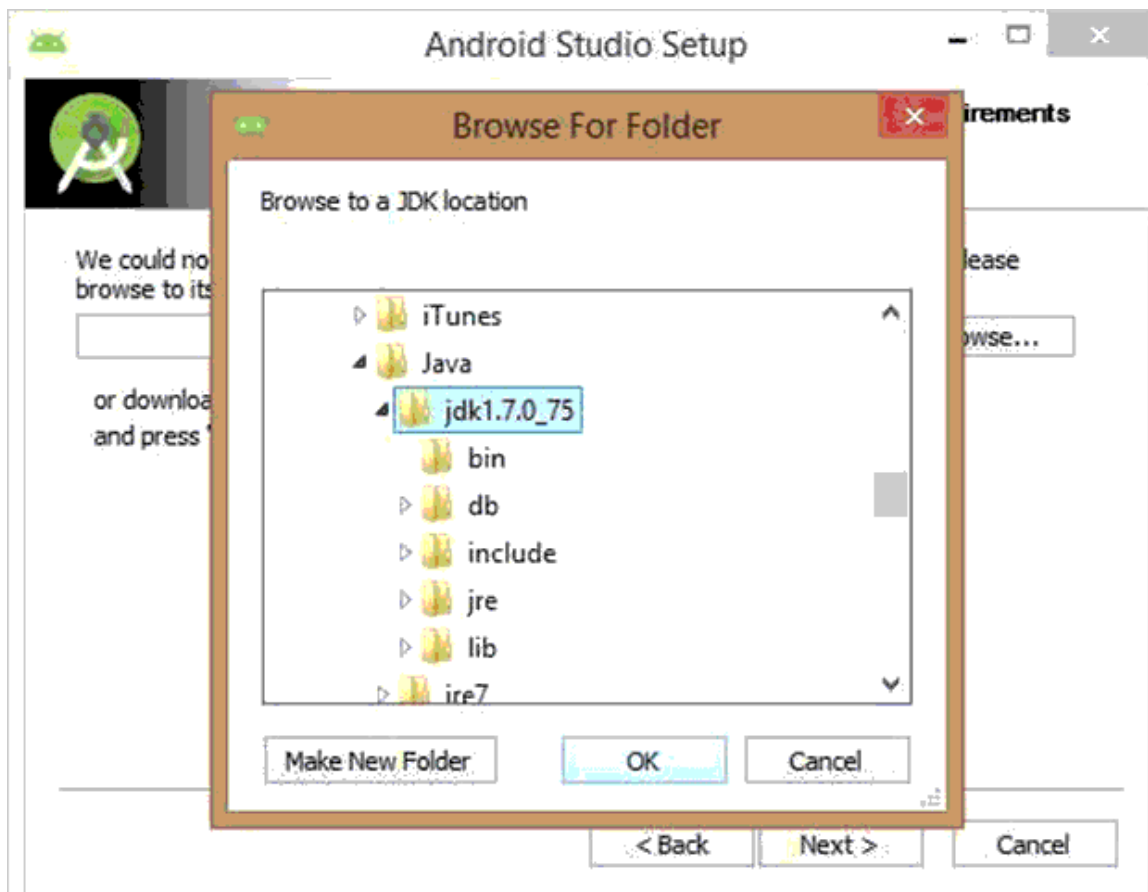
After download Android Studio, run its setup and install it. To install follow the instructions.



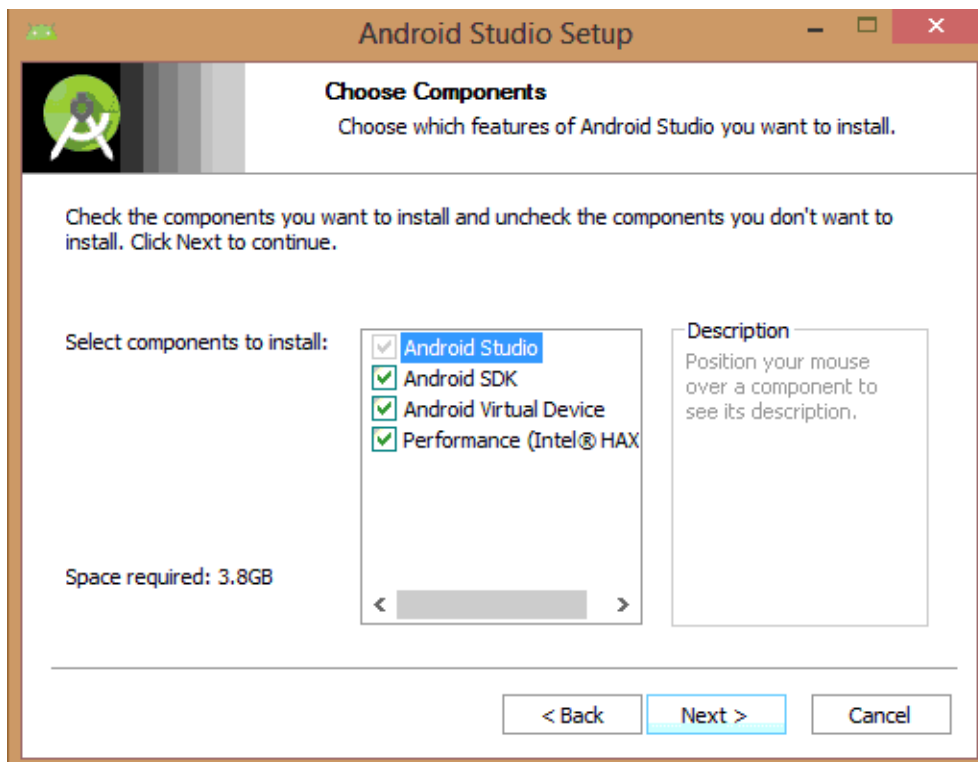
Click Next to proceed.



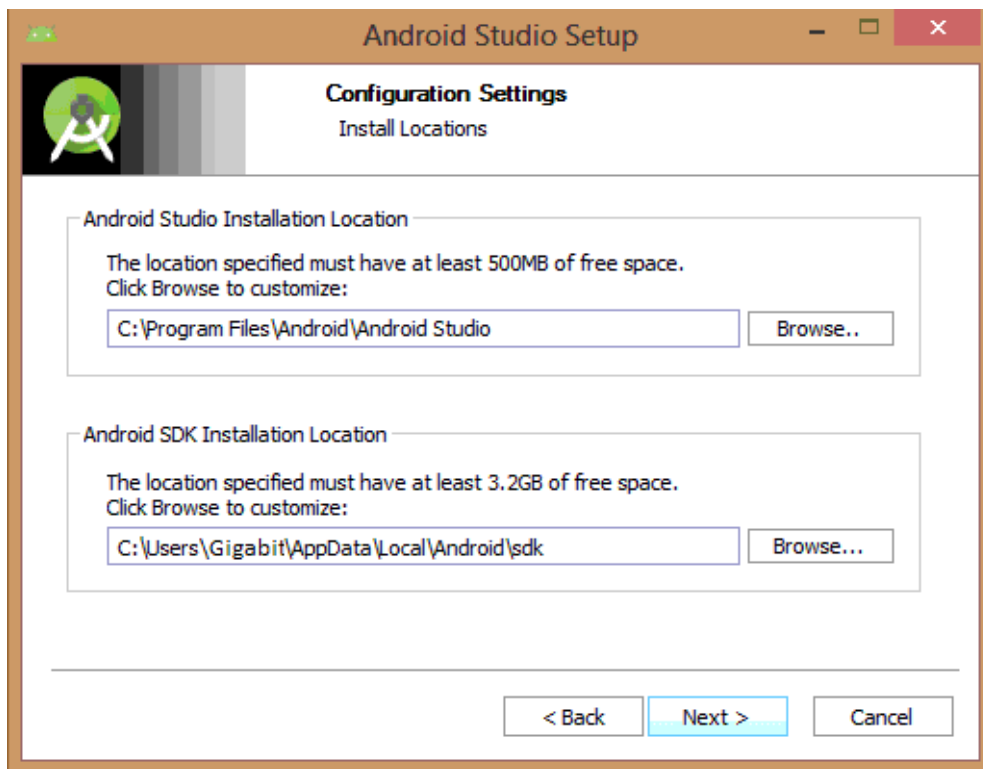
Browse JDK path and click Next



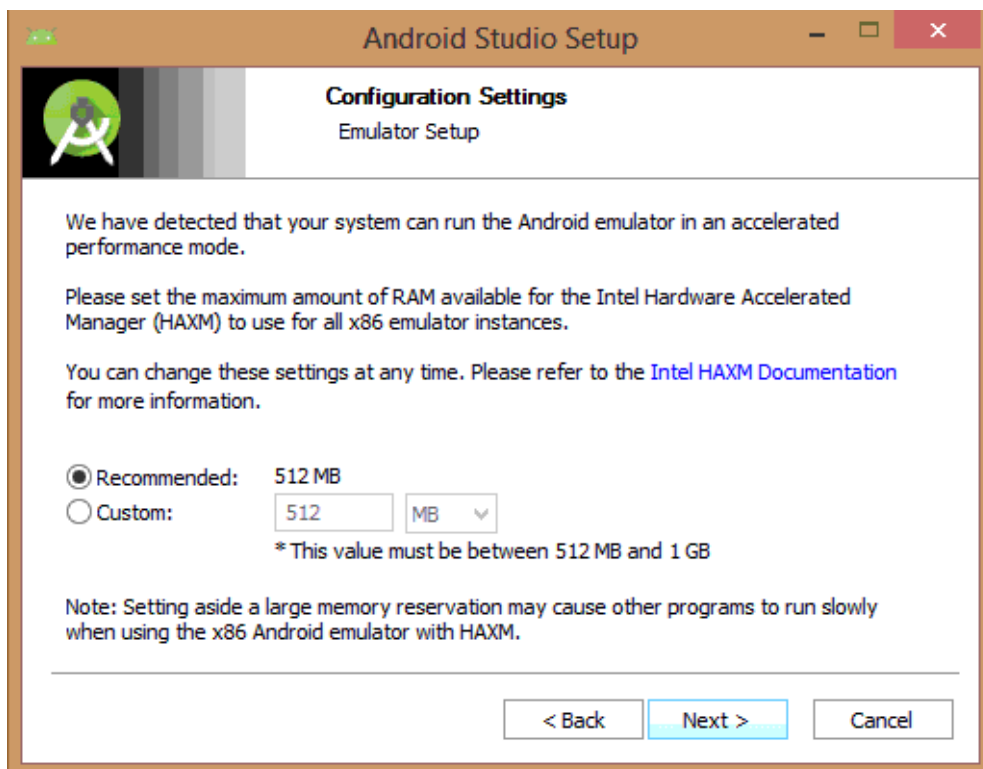
Check the component which we want to install.



Provide Android SDK installation location.

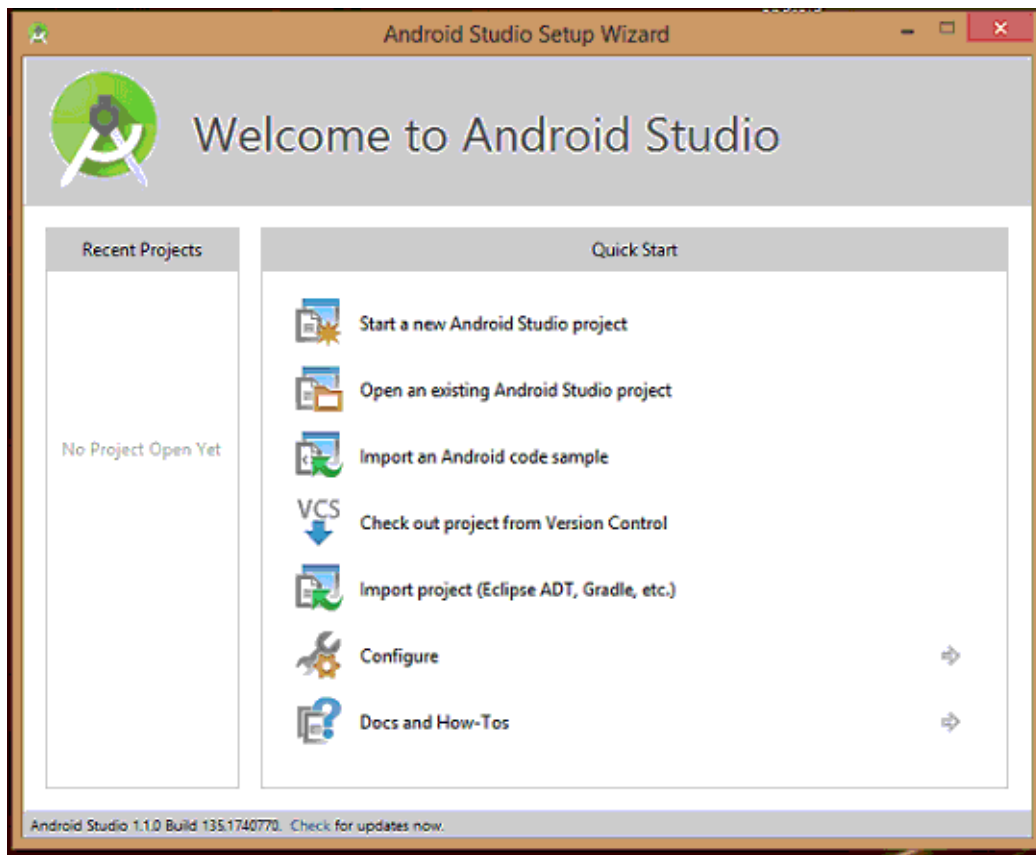


Select the maximum size of RAM to use emulator instances.



Android Studio is successfully installed on our system.

OUTPUT:



EX. NO: 2 DEVELOP AN APPLICATION THAT USES WIDGETS, GUI COMPONENTS, FONTS, AND COLORS

Java file

```
package com.lab.guicomponents;

import android.os.Bundle;
import android.app.Activity;
import android.view.Menu;

public class MainActivity extends Activity {

    @Override

    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    @Override

    public boolean onCreateOptionsMenu(Menu menu) {
        // Inflate the menu; this adds items to the action bar if it is present.
        getMenuInflater().inflate(R.menu.main, menu);
        return true;
    }

}
```

XML file

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent"
    android:orientation="vertical"
    android:padding="20dp">
```

```

<EditText
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Edit Text"/>

<Button
    android:id="@+id/btn1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Button"/>

<CheckBox
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Checkbox"/>

<RadioButton
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Radio Button"/>

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Green Color"
    android:textColor="#00ff00"/>

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Red Color"
    android:textColor="#ff0000"/>

<TextView
    android:layout_width="wrap_content"

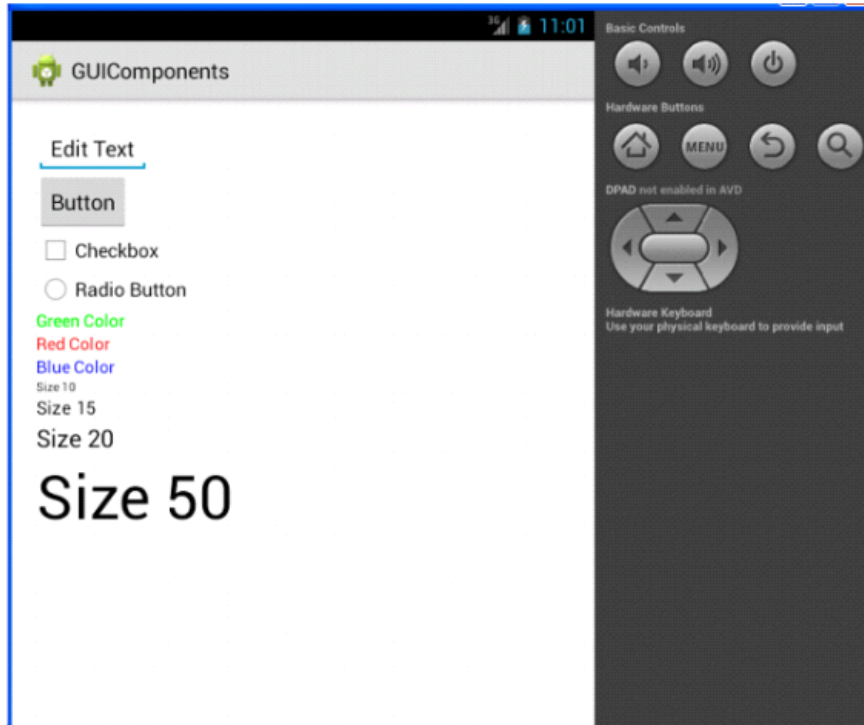
```



```
    android:layout_height="wrap_content"
    android:text="Blue Color"
    android:textColor="#0000ff"/>
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Size 10"
    android:textColor="#000000"
    android:textSize="10dip"/>
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"

    android:text="Size 15"
    android:textColor="#000000"
    android:textSize="15dip"/>
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Size 20"
    android:textColor="#000000"
    android:textSize="20dip"/>
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Size 50"
    android:textColor="#000000"
    android:textSize="50dip"/>
</LinearLayout>
```

OUTPUT:



EX NO: 3 **DEVELOP A NATIVE CALCULATOR APPLICATION**

JAVA CODE

```
package com.example.rating;

import androidx.appcompat.app.AppCompatActivity;
import android.annotation.SuppressLint;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
    EditText firstnum,secondnum;
    TextView r;
    Button bt;
    Button bts;
    Button btm;
    double a,b,c;
    @SuppressWarnings("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        firstnum=(EditText) findViewById(R.id.first);
        secondnum=(EditText) findViewById(R.id.second);
        bt=(Button) findViewById(R.id.buttonadd);
        bts=(Button) findViewById(R.id.buttonsub);
        btm=(Button) findViewById(R.id.buttonmul);
        r=(TextView) findViewById(R.id.result);
```

```

bt.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v) {
a=Double.parseDouble(firstnum.getText().toString());
b=Double.parseDouble(secondnum.getText().toString());
c=a+b;
r.setText("Sum="+c);
}
});

bts.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v) {
a=Double.parseDouble(firstnum.getText().toString());
b=Double.parseDouble(secondnum.getText().toString());
c=a-b;
r.setText("Sub="+c);
}
});

btm.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v) {
a=Double.parseDouble(firstnum.getText().toString());
b=Double.parseDouble(secondnum.getText().toString());
c=a*b;
r.setText("Mul="+c);
}
});
}
}
}

```

XML CODE

```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:layout_centerHorizontal="true"
tools:context=".MainActivity">
<TextView
android:id="@+id/textview1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignParentTop="true"
android:layout_centerHorizontal="true"
android:text="Enter number 1"
android:textSize="18sp" />
<EditText
android:id="@+id/first"
android:text="one"
android:minHeight="48dp"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_below="@id/textview1"
android:layout_alignParentTop="false"
android:layout_alignParentRight="false"
android:layout_centerHorizontal="true"
android:ems="10"
android:inputType="number" />
```

```

<TextView
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/first"
    android:layout_centerHorizontal="true"
    android:text="Enter Number 2"
    android:textSize="18sp"/>

<EditText
    android:id="@+id/second"
    android:text="two"
    android:minHeight="48dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/textView2"
    android:layout_centerHorizontal="true"
    android:ems="10"
    android:inputType="number" />

<Button
    android:id="@+id/buttonadd"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/second"
    android:layout_centerHorizontal="true"
    android:text="Add"/>

<Button
    android:id="@+id/buttonsub"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"

```

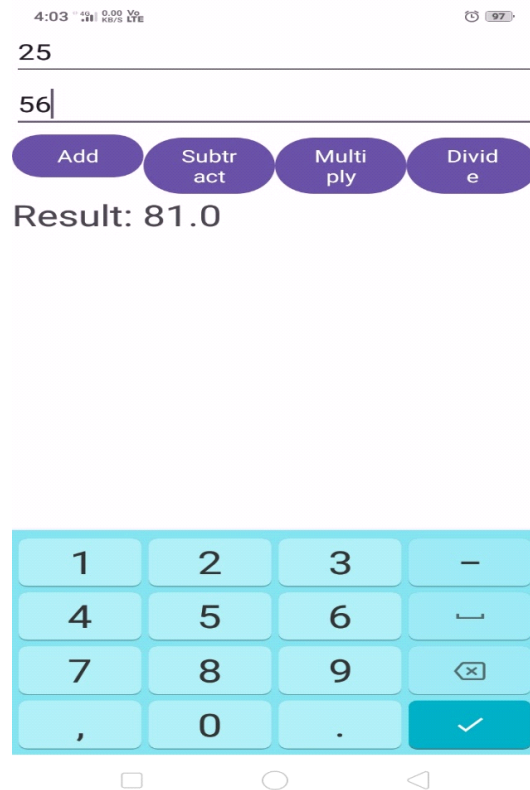
```
    android:layout_below="@id/buttonadd"
    android:layout_centerHorizontal="true"
    android:text="Sub" />

    <Button
    android:id="@+id/buttonmul"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/buttonsub"
    android:layout_centerHorizontal="true"
    android:text="mul" />

    <TextView
    android:id="@+id/result"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/buttonmul"
    android:layout_centerHorizontal="true"
    android:textSize="24sp" />

</RelativeLayout>
```

OUTPUT:



EX NO: 4 **DEVELOP A GAMING APPLICATION THAT USES 2-D ANIMATIONS AND GESTURES**

JAVA CODE

```
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.view.animation.Animation;
import android.view.animation.AnimationUtils;
import android.widget.Button;
import android.widget.ImageView;

public class MainActivity extends AppCompatActivity {
    ImageView imageView;

    Button blinkBTN, rotateBTN, fadeBTN, moveBTN, slideBTN, zoomBTN,
    stopBTN;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        imageView = findViewById(R.id.imageview);
        blinkBTN = findViewById(R.id.BTNblink);
        rotateBTN = findViewById(R.id.BTNrotate);
        fadeBTN = findViewById(R.id.BTNfade);
        moveBTN = findViewById(R.id.BTNmove);
        slideBTN = findViewById(R.id.BTNslide);
        zoomBTN = findViewById(R.id.BTNzoom);
        stopBTN = findViewById(R.id.BTNstop);

        blinkBTN.setOnClickListener(new View.OnClickListener() {
```

```

        @Override
        public void onClick(View v) {
            // To add blink animation
            Animation animation =
AnimationUtils.loadAnimation(getApplicationContext(), R.anim.blink_animation);
            imageView.startAnimation(animation);
        }
    });

    rotateBTN.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            // To add rotate animation
            Animation animation =
AnimationUtils.loadAnimation(getApplicationContext(), R.anim.rotate_animation);
            imageView.startAnimation(animation);
        }
    });

    fadeBTN.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            // To add fade animation
            Animation animation =
AnimationUtils.loadAnimation(getApplicationContext(), R.anim.fade_animation);
            imageView.startAnimation(animation);
        }
    });

    moveBTN.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            // To add move animation

```

```

        Animation animation =
AnimationUtils.loadAnimation(getApplicationContext(), R.anim.move_animation);

        imageView.startAnimation(animation);

    }

});

slideBTN.setOnClickListener(new View.OnClickListener() {

    @Override

    public void onClick(View v) {

        // To add slide animation

        Animation animation =
AnimationUtils.loadAnimation(getApplicationContext(), R.anim.slide_animation);

        imageView.startAnimation(animation);

    }

});

zoomBTN.setOnClickListener(new View.OnClickListener() {

    @Override

    public void onClick(View v) {

        Animation animation =
AnimationUtils.loadAnimation(getApplicationContext(), R.anim.zoom_animation);

        imageView.startAnimation(animation);

    }

});

stopBTN.setOnClickListener(new View.OnClickListener() {

    @Override

    public void onClick(View v) {

        imageView.clearAnimation();

    }

});

}

}

```

XML CODE

```
<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout

    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ImageView

        android:id="@+id/imageview"
        android:layout_width="200dp"
        android:layout_height="200dp"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="40dp"
        android:contentDescription="@string/app_name"
        android:src="@drawable/gfgimage" />

    <LinearLayout

        android:id="@+id/linear1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_below="@id/imageview"
        android:layout_marginTop="30dp"
        android:orientation="horizontal"
        android:weightSum="3">

        <!--To start the blink animation of the image-->

        <Button

            android:id="@+id/BTNblink"

            style="@style/TextAppearance.AppCompat.Widget.Button"
            android:layout_width="0dp"
```

```

        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:layout_weight="1"
        android:padding="3dp"
        android:text="@string/blink"
        android:textColor="@color/white" />
<!--To start the rotate animation of the image-->
<Button
    android:id="@+id/BTNrotate"
    style="@style/TextAppearance.AppCompat.Widget.Button"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_margin="10dp"
    android:layout_weight="1"
    android:padding="3dp"
    android:text="@string/clockwise"
    android:textColor="@color/white" />
<!--To start the fading animation of the image-->
<Button
    android:id="@+id/BTNfade"
    style="@style/TextAppearance.AppCompat.Widget.Button"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_margin="10dp"
    android:layout_weight="1"
    android:padding="3dp"
    android:text="@string/fade"
    android:textColor="@color/white" />
</LinearLayout>

```

<LinearLayout

```
    android:id="@+id/linear2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@id/linear1"
    android:layout_marginTop="30dp"
    android:orientation="horizontal"
    android:weightSum="3">
    <!--To start the move animation of the image-->
```

<Button

```
    android:id="@+id/BTNmove"
    style="@style/TextAppearance.AppCompat.Widget.Button"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_margin="10dp"
    android:layout_weight="1"
    android:padding="3dp"
    android:text="@string/move"
    android:textColor="@color/white" />
```

<!--To start the slide animation of the image-->

<Button

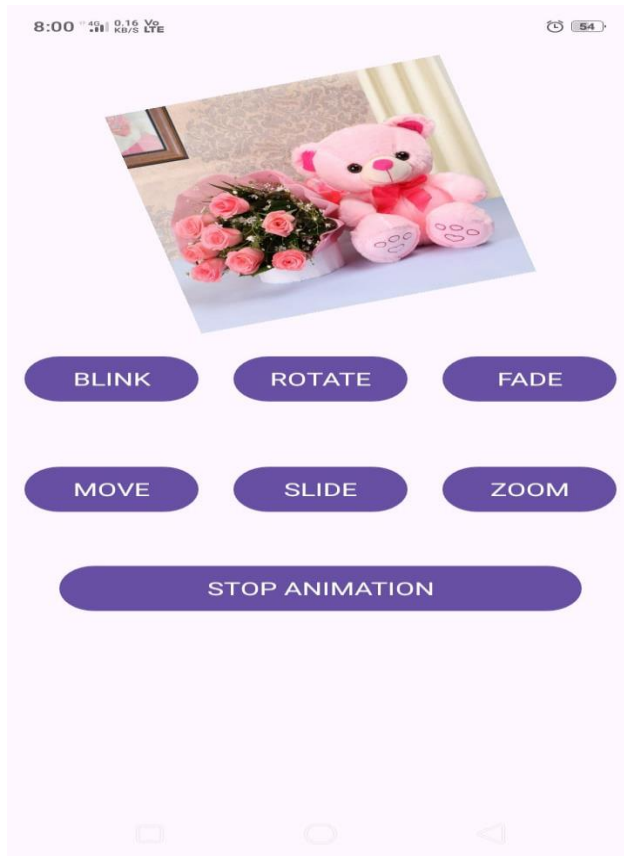
```
    android:id="@+id/BTNslide"
    style="@style/TextAppearance.AppCompat.Widget.Button"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_margin="10dp"
    android:layout_weight="1"
    android:padding="3dp"
    android:text="@string/slide"
```

```

        android:textColor="@color/white" />
<!--To start the zoom animation of the image-->
<Button
    android:id="@+id/BTNzoom"
    style="@style/TextAppearance.AppCompat.Widget.Button"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_margin="10dp"
    android:layout_weight="1"
    android:padding="3dp"
    android:text="@string/zoom"
    android:textColor="@color/white" />
</LinearLayout>
<!--To stop the animation of the image-->
<Button
    android:id="@+id/BTNstop"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@id/linear2"
    android:layout_marginLeft="30dp"
    android:layout_marginTop="30dp"
    android:layout_marginRight="30dp"
    android:text="@string/stop_animation" />
</RelativeLayout>

```

OUTPUT:



EX NO: 5 **DEVELOP A MOVIE RATING APPLICATION (SIMILAR TO IMDB)**

JAVA CODE

```
package com.example.ra;

import android.graphics.Color;
import android.graphics.PorterDuff;
import android.graphics.drawable.LayerDrawable;
import androidx.appcompat.app.AppCompatActivity;
//import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.RatingBar;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    RatingBar rt;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        //binding MainActivity.java with activity_main.xml file
        rt = (RatingBar) findViewById(R.id.ratingBar);

        //finding the specific RatingBar with its unique ID
        LayerDrawable stars=(LayerDrawable)rt.getProgressDrawable();

        //Use for changing the color of RatingBar
        stars.getDrawable(2).setColorFilter(Color.YELLOW,
        PorterDuff.Mode.SRC_ATOP);
    }

    public void Call(View v)
```

```

{
// This function is called when button is clicked.
// Display ratings, which is required to be converted into string
first.
TextView t = (TextView)findViewById(R.id.textView2);
t.setText("You Rated :"+String.valueOf(rt.getRating()));
}
}

```

XML CODE

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:fitsSystemWindows="true"
tools:context=".MainActivity">

<RatingBar
android:id="@+id/ratingBar"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="104dp"
android:background="#00AA00"
app:layout_constraintLeft_toLeftOf="parent"
app:layout_constraintRight_toRightOf="parent"
app:layout_constraintTop_toTopOf="parent"
tools:layout_constraintLeft_creator="1"

```

```

tools:layout_constraintRight_creator="1"
tools:layout_constraintTop_creator="1" />
<TextView
    android:id="@+id/textView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Rate Me!!!"
    android:textColor="@android:color/background_dark"
    android:textSize="30sp"
    android:textStyle="bold|italic"
    tools:layout_editor_absoluteX="127dp"
    tools:layout_editor_absoluteY="28dp" />
<TextView
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="148dp"
    android:textColorHint="#FF0000"
    android:textSize="24sp"
    app:layout_constraintLeft_toLeftOf="parent"
    app:layout_constraintRight_toRightOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/ratingBar"
    tools:layout_constraintRight_creator="1"
    tools:layout_constraintLeft_creator="1" />
<Button
    android:id="@+id/button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginBottom="50dp"

```

```
android:layout_marginTop="50dp"
android:background="@android:color/holo_red_dark"
android:onClick="Call"
android:text="Submit"
android:textColor="@android:color/background_light"
android:textStyle="bold|italic"
app:layout_constraintBottom_toTopOf="@+id/textView2"
app:layout_constraintLeft_toLeftOf="parent"
app:layout_constraintRight_toRightOf="parent"
app:layout_constraintTop_toBottomOf="@+id/ratingBar"
tools:layout_constraintBottom_creator="1"
tools:layout_constraintLeft_creator="1"
tools:layout_constraintRight_creator="1"
tools:layout_constraintTop_creator="1" />

</androidx.coordinatorlayout.widget.CoordinatorLayout>
```

OUTPUT:

OUTPUT



EX NO: 6 DEVELOP AN APPLICATION TO CONNECT TO A WEB SERVICE AND TO RETRIEVE DATA WITH HTTP

JAVA CODE:

```
package com.example.webv;  
import android.os.Bundle;  
import android.webkit.WebView;  
import android.webkit.WebViewClient;  
import androidx.appcompat.app.AppCompatActivity;  
public class MainActivity extends AppCompatActivity {  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.activity_main);  
        // Find the WebView by its unique ID  
        WebView webView = findViewById(R.id.web);  
        // loading https://www.geeksforgeeks.org url in the WebView.  
        webView.loadUrl("https://www.geeksforgeeks.org");  
        // this will enable the javascript.  
        webView.getSettings().setJavaScriptEnabled(true);  
        // WebViewClient allows you to handle  
        // onPageFinished and override Url loading.  
        webView.setWebViewClient(new WebViewClient());  
    }  
}
```

XML CODE:

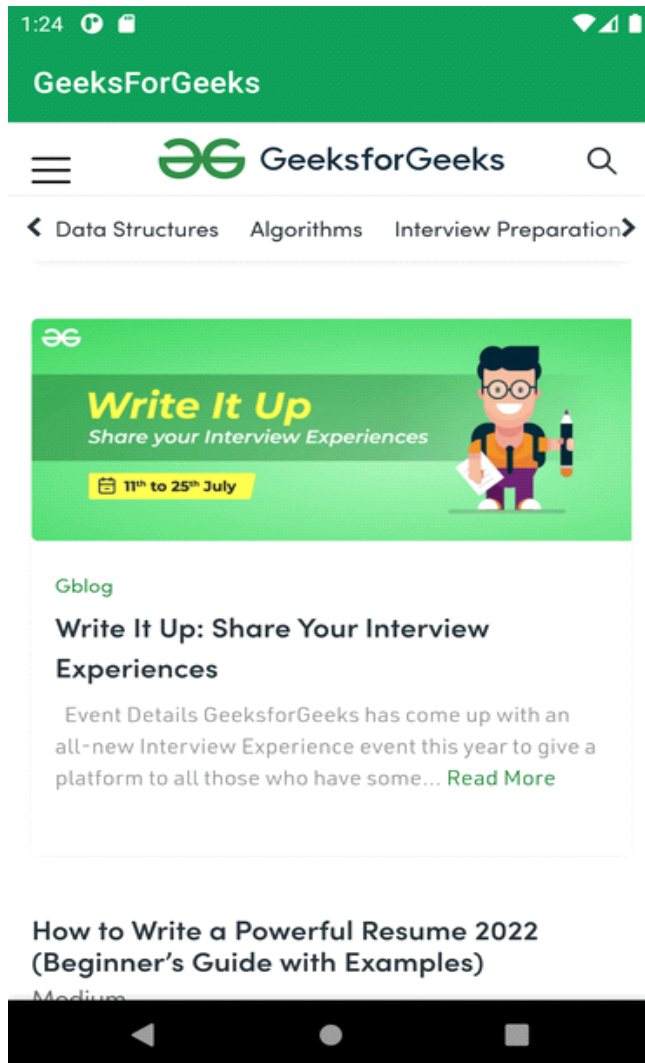
```
<?xml version="1.0" encoding="utf-8"?>  
<RelativeLayout  
    xmlns:android="http://schemas.android.com/apk/res/android"
```

```
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context="com.gfg.android.examplekotlin.MainActivity">
<!-- unique ID of WebView -- >
<WebView
android:id="@+id/web"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:layout_editor_absoluteX="8dp"
tools:layout_editor_absoluteY="8dp" />
</RelativeLayout>
```

AndroidManifest.xml

```
<uses-permission android:name="android.permission.INTERNET" />
```

OUTPUT:



EX NO: 7 DEVELOP A SIMPLE SHOPPING APPLICATION

JAVA CODE:

```
package com.example.grocerylist.Adapter;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.recyclerview.widget.RecyclerView;
import com.example.grocerylist.Database.Entity.GroceryItems;
import com.example.grocerylist.R;
import com.example.grocerylist.UI.GroceryViewModel;
import com.example.grocerylist.databinding.GroceryadapterBinding;

public class GroceryAdapter extends
RecyclerView.Adapter<GroceryAdapter.GroceryViewHolder> {
    private List<GroceryItems> list;
    private GroceryViewModel viewModel;

    public GroceryAdapter(List<GroceryItems> list, GroceryViewModel viewModel) {
        this.list = list;
        this.viewModel = viewModel;
    }

    @Override
    public GroceryViewHolder onCreateViewHolder(ViewGroup parent, int viewType)
    {
        GroceryadapterBinding binding =
GroceryadapterBinding.inflate(LayoutInflater.from(parent.getContext()), parent,
false);
```

```

        return new GroceryViewHolder(binding);
    }

    @Override
    public int getItemCount() {
        return list.size();
    }

    @Override
    public void onBindViewHolder(GroceryViewHolder holder, int position) {
        GroceryItems currentPosition = list.get(position);
        holder.binding.txtItemName.setText(currentPosition.getItemName());

        holder.binding.txtItemPrice.setText(String.valueOf(currentPosition.getItemPrice()));

        holder.binding.txtItemQuantity.setText(String.valueOf(currentPosition.getItemQuantity()));

        holder.binding.btnDelete.setOnClickListener(v ->
            viewModel.delete(currentPosition));

        if (position == list.size() - 1) {
            int totalCost = 0;
            for (int i = 0; i < list.size(); i++) {
                totalCost += list.get(i).getItemPrice();
            }
            holder.binding.txtItemTotalCost.setVisibility(View.VISIBLE);
            holder.binding.txtTotalCostTitle.setVisibility(View.VISIBLE);
            holder.binding.txtItemTotalCost.setText(String.valueOf(totalCost));
        }
    }
}

```

```

static class GroceryViewHolder extends RecyclerView.ViewHolder {
    private final GroceryadapterBinding binding;

    public GroceryViewHolder(GroceryadapterBinding binding) {
        super(binding.getRoot());
        this.binding = binding;
    }
}
}

```

XML CODE:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="125dp"
    android:background="@drawable/adapter1">

    <!-- To display item name in recycler view -->
    <TextView
        android:id="@+id/txtItemName"
        android:layout_width="0dp"
        android:layout_height="53dp"
        android:layout_marginTop="16dp"
        android:layout_marginEnd="15dp"
        android:layout_marginRight="15dp"
        android:layout_marginBottom="17dp"

```

```
android:fontFamily="@font/rokkitt"
android:text="@string/itemName"
android:textColor="@color/white"
android:textSize="35sp"
app:layout_constraintBottom_toTopOf="@+id/txtTotalCostTitle"
app:layout_constraintEnd_toStartOf="@+id/txtItemQuantity"
app:layout_constraintStart_toEndOf="@+id/cbItemCheck"
app:layout_constraintTop_toTopOf="parent" />
```

<!-- To display item quantity -->

<TextView

```
android:id="@+id/txtItemQuantity"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginEnd="34dp"
android:layout_marginRight="34dp"
android:layout_marginBottom="9dp"
android:text="@string/itemQuantity"
android:textColor="@color/white"
android:textSize="25sp"
app:layout_constraintBottom_toBottomOf="@+id/txtItemName"
app:layout_constraintEnd_toStartOf="@+id/txtItemPrice"
app:layout_constraintStart_toEndOf="@+id/txtItemName"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="1.0" />
```

<!-- To display item price -->

<TextView

```
android:id="@+id/txtItemPrice"
```

```

android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="26dp"
android:layout_marginEnd="22dp"
android:layout_marginRight="22dp"
android:layout_marginBottom="26dp"
android:text="@string/itemPrice"
android:textColor="@color/white"
android:textSize="25sp"
android:textStyle="bold"
app:layout_constraintBottom_toTopOf="@+id/txtItemTotalCost"
app:layout_constraintEnd_toStartOf="@+id/ibDelete"
app:layout_constraintStart_toEndOf="@+id/txtItemQuantity"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.0" />

```

<CheckBox

```

android:id="@+id/cbItemCheck"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginStart="29dp"
android:layout_marginLeft="29dp"
android:layout_marginEnd="16dp"
android:layout_marginRight="16dp"
android:background="@color/white"
android:shadowColor="@color/black"
app:layout_constraintBaseline_toBaselineOf="@+id/txtItemName"
app:layout_constraintEnd_toStartOf="@+id/txtItemName"
app:layout_constraintStart_toStartOf="parent" />

```

<!-- This button is used to delete grocery item -->

<ImageButton

```
    android:id="@+id/ibDelete"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginEnd="26dp"
    android:layout_marginRight="26dp"
    android:background="@color/black"
    android:src="@drawable/ic_action_delete"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toEndOf="@+id/txtItemPrice"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.257" />
```

<!-- To display total cost of grocery items -->

<TextView

```
    android:id="@+id/txtItemTotalCost"
    android:layout_width="100dp"
    android:layout_height="60dp"
    android:background="@drawable/adapter2"
    android:padding="8dp"
    android:paddingLeft="12dp"
    android:text="@string/totalCost"
    android:textSize="30sp"
    android:textStyle="bold"
    android:visibility="invisible"
    app:layout_constraintBottom_toBottomOf="parent"
```

```
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintStart_toEndOf="@+id/txtTotalCostTitle"
app:layout_constraintTop_toTopOf="@+id/txtTotalCostTitle"
app:layout_constraintVertical_bias="0.875" />
```

<!-- This text view is used to add statement for total cost -->

<TextView

```
    android:id="@+id/txtTotalCostTitle"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="77dp"
    android:layout_marginLeft="77dp"
    android:layout_marginEnd="149dp"
    android:layout_marginRight="149dp"
    android:layout_marginBottom="16dp"
    android:text="@string/totalCostTitle"
    android:textColor="@color/white"
    android:textSize="20dp"
    android:visibility="invisible"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toStartOf="@+id/txtItemTotalCost"
    app:layout_constraintStart_toStartOf="parent" />
```

</androidx.constraintlayout.widget.ConstraintLayout>

OUTPUT:

Two side-by-side screenshots of a web form titled "Add Items to your cart".

The left screenshot shows the form with a light green header and a white body. The form contains a table with the following structure:

Item	Quantity	Price
Cancel Save		

The right screenshot shows the same form with a dark teal background. The form contains a table with the following structure:

Item Name	Item Quantity	Item Price
Cancel Save		

EX NO: 8 DESIGN A WEB SERVER SUPPORTING PUSH NOTIFICATIONS

JAVA CODE:

```
package com.example.notification;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;
import android.app.NotificationManager;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {
    Button button;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        button=findViewById(R.id.click);
        button.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                NotificationCompat.Builder
                mbuilder=(NotificationCompat.Builder)
                new NotificationCompat.Builder(getApplicationContext())
                .setSmallIcon(R.drawable.one)
                .setContentTitle("Notification")
                .setContentText("this is noti");
                NotificationManager notificationManager=(NotificationManager)
                getSystemService(NOTIFICATION_SERVICE);
```

```

notificationManager.notify(0,mbuilder.build());
}
});

}
}

```

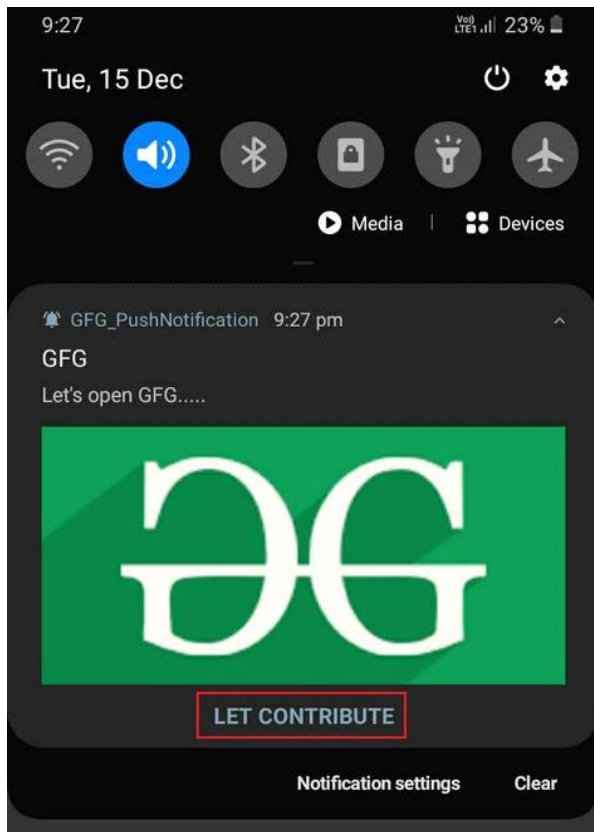
XML:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
xmlns:app="http://schemas.android.com/apk/res-auto"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
<Button
android:id="@+id/click"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:contentDescription="button"
android:layout_centerInParent="true"
android:text="notification"
/>
</RelativeLayout>

```

OUTPUT:



EX NO: 9 DEVELOP AN APPLICATION BY INTEGRATING GOOGLE MAPS

JAVA CODE:

```
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import com.google.android.gms.maps.CameraUpdateFactory;
import com.google.android.gms.maps.GoogleMap;
import com.google.android.gms.maps.OnMapReadyCallback;
import com.google.android.gms.maps.SupportMapFragment;
import com.google.android.gms.maps.model.LatLng;
import com.google.android.gms.maps.model.MarkerOptions;

public class MainActivity extends AppCompatActivity implements
OnMapReadyCallback {
    private GoogleMap mMap;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        SupportMapFragment mapFragment = (SupportMapFragment)
getSupportFragmentManager().findFragmentById(R.id.map);
        mapFragment.getMapAsync(this);
    }
    @Override
    public void onMapReady(GoogleMap googleMap) {
        mMap = googleMap;

        // Add a marker in Sydney and move the camera
        LatLng mayiladuthurai = new LatLng(11.1035, 79.6521);
        mMap.addMarker(new MarkerOptions().position(mayiladuthurai).title("Marker in
Mayiladuthurai"));
        mMap.moveCamera(CameraUpdateFactory.newLatLng(mayiladuthurai));

    }
}
```

XML CODE:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <fragment
        android:id="@+id/map"
        android:name="com.google.android.gms.maps.SupportMapFragment"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</RelativeLayout>
```

ANDROID MANIFEST CODE:

The below code must be in application tag

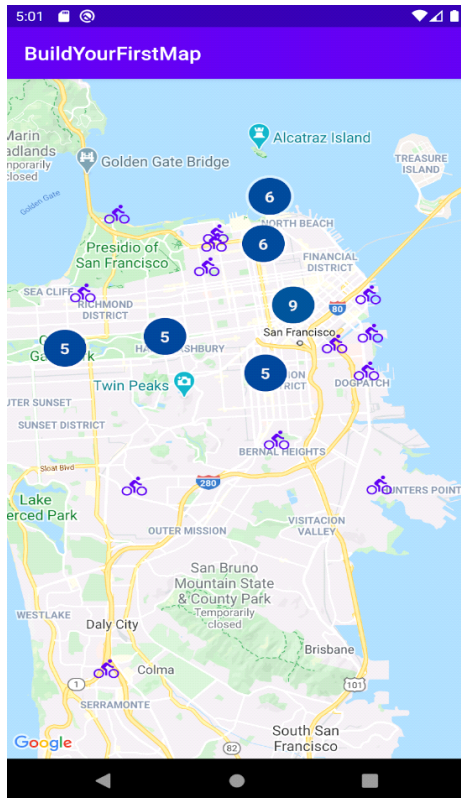
```
<meta-data
    android:name="com.google.android.geo.API_KEY"
    android:value="// your API "/>
```

Build.gradle.kts(Module:app)

The code must be in dependencies tag

```
implementation("com.google.android.gms:play-services-maps:18.1.0")
```

OUTPUT:



JAVA CODE:

```
package com.example.db;
import android.app.Activity;
import android.app.AlertDialog.Builder;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;
import android.widget.EditText;
public class MainActivity extends Activity implements OnClickListener {
    EditText Rollno, Name, Marks;
    Button Insert, Delete, Update, View, ViewAll;
    SQLiteDatabase db;

    /**
     * Called when the activity is first created.
     */
    @Override
    public void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);
        Rollno = (EditText) findViewById(R.id.Rollno);
        Name = (EditText) findViewById(R.id.Name);
        Marks = (EditText) findViewById(R.id.Marks);
        Insert = (Button) findViewById(R.id.Insert);
        Delete = (Button) findViewById(R.id.Delete);
        Update = (Button) findViewById(R.id.Update);
        View = (Button) findViewById(R.id.View);
        ViewAll = (Button) findViewById(R.id.all);
        Insert.setOnClickListener(this);
        Delete.setOnClickListener(this);
        Update.setOnClickListener(this);
        View.setOnClickListener(this);
        ViewAll.setOnClickListener(this);
        // Creating database and table
        db = openOrCreateDatabase("StudentDB", Context.MODE_PRIVATE, null);
        db.execSQL("CREAT TABLE IF NOT EXISTS student(rollno VARCHAR,name
        VARCHAR, marks VARCHAR);");
    }
}
```

```

public void onClick(View view) {
    // Inserting a record to the Student table
    if (view == Insert) {
        // Checking for empty fields
        if (Rollno.getText().toString().trim().length() == 0 ||
            Name.getText().toString().trim().length() == 0 ||
            Marks.getText().toString().trim().length() == 0) {
            showMessage("Error", "Please enter all values");
            return;
        }
        db.execSQL("INSERT INTO studet VALUES('"+ Rollno.getText() +
            "','"+ Name.getText() + "','"+ Marks.getText() + "','");");
        showMessage("Success", "Record added");
        clearText();
    }
    // Deleting a record from the Student table
    if (view == Delete) {
        // Checking for empty roll number
        if (Rollno.getText().toString().trim().length() == 0) {
            showMessage("Error", "Please enter Rollno");
            return;
        }
        Cursor c = db.rawQuery("SELECT * FROM student WHERE rollno='"+
Rollno.getText() + "'", null);
        if (c.moveToFirst()) {
            db.execSQL("DELETE FROM student WHERE rollno='"+ Rollno.getText() +
            "'");
            showMessage("Success", "Record Deleted");
        } else {
            showMessage("Error", "Invalid Rollno");
        }
        clearText();
    }
    // Updating a record in the Student table
    if (view == Update) {
        // Checking for empty roll number
        if (Rollno.getText().toString().trim().length() == 0) {
            showMessage("Error", "Please enter Rollno");
            return;
        }
        Cursor c = db.rawQuery("SELECT * FROM student WHERE rollno='"+
Rollno.getText() + "'", null);
        if (c.moveToFirst()) {
            db.execSQL("UPDATE student SET name='"+ Name.getText() + "','marks='"+
Marks.getText() + "' WHERE rollno='"+ Rollno.getText() + "'");

```



```

        showMessage("Success", "Record Modified");
    } else {
        showMessage("Error", "Invalid Rollno");
    }
    clearText();
}
// Display a record from the Student table
if (view == View)

    // Checking for empty roll number
    if (Rollno.getText().toString().trim().length() == 0) {
        showMessage("Error", "Please enter Rollno");
        return;
    }
    Cursor c = db.rawQuery("SELECT * FROM student WHERE rollno=" +
Rollno.getText() + "", null);
    if (c.moveToFirst()) {
        Name.setText(c.getString(1));
        Marks.setText(c.getString(2));
    } else {
        showMessage("Error", "Invalid Rollno");
    }
    clearText();
}

if(view == ViewAll)
{
    c = db.rawQuery("SELECT * FROM student", null);
    if(c.getCount()==0)
    {
        showMessage("Error", "No records found");
        return;
    }
    StringBuffer buffer=new StringBuffer();
    while(c.moveToNext())
    {
        buffer.append("Rollno: "+c.getString(0)+"\n");
        buffer.append("Name: "+c.getString(1)+"\n");
        buffer.append("Marks: "+c.getString(2)+"\n\n");
    }
    showMessage("Student Details", buffer.toString());
}
}

public void showMessage(String title,String message)
{
    Builder builder=new Builder(this);
    builder.setCancelable(true);

```

```

        builder.setTitle(title);
        builder.setMessage(message);
        builder.show();
    }
    public void clearText()
    {
        Rollno.setText("");
        Name.setText("");
        Marks.setText("");
        Rollno.requestFocus();
    }
}

```

XML CODE

```

<?xml version="1.0" encoding="utf-8"?>
<AbsoluteLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="50dp"
        android:layout_y="20dp"
        android:text="Student Details"
        android:textSize="30sp" />
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="20dp"
        android:layout_y="110dp"
        android:text="Enter Rollno:"
        android:textSize="20sp" />
    <EditText
        android:id="@+id/Rollno"
        android:layout_width="150dp"
        android:layout_height="wrap_content"
        android:layout_x="175dp"
        android:layout_y="100dp"
        android:inputType="number"
        android:textSize="20sp"
        android:text="rollno"

    />
    <TextView
        android:layout_width="wrap_content"

```

```

        android:layout_height="wrap_content"
        android:layout_x="20dp"
        android:layout_y="160dp"
        android:text="Enter Name:"
        android:textSize="20sp" />
<EditText
    android:id="@+id/Name"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="175dp"
    android:layout_y="150dp"
    android:inputType="text"
    android:text="Enter Name:"
    android:textSize="20sp" />
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_x="20dp"
    android:layout_y="210dp"
    android:text="Enter Marks:"
    android:textSize="20sp" />
<EditText
    android:id="@+id/Marks"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:text="Enter Mark:"
    android:layout_x="175dp"
    android:layout_y="200dp"
    android:inputType="number"
    android:textSize="20sp" />
<Button
    android:id="@+id/Insert"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="25dp"
    android:layout_y="300dp"
    android:text="Insert"
    android:textSize="30dp" />
<Button
    android:id="@+id/Delete"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="200dp"
    android:layout_y="300dp"
    android:text="Delete"
    android:textSize="30dp" />

```

```
<Button
    android:id="@+id/Update"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="25dp"
    android:layout_y="400dp"
    android:text="Update"
    android:textSize="30dp" />
<Button
    android:id="@+id/View"
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:layout_x="200dp"
    android:layout_y="400dp"
    android:text="View"
    android:textSize="30dp" />

<Button
    android:id="@+id/all"
    android:layout_width="200dp"
    android:layout_height="wrap_content"
    android:layout_x="100dp"
    android:layout_y="500dp"
    android:text="all"
    android:textSize="30dp" />
</AbsoluteLayout>
```

OUTPUT:

6:40 5G 100% 5G

Student Details

Enter Rollno:

Enter Name:

Enter Marks:

INSERT **DELETE**

UPDATE **VIEW**

1 2 3 -

4 5 6 ↵

7 8 9 ✕

, 0 . →

6:39 5G 100% 5G

Student Details

Enter Rollno:

Student Details

Rollno: 1001
Name: kani
Marks: 100

Rollno: 1002
Name: rani
Marks: 99

Rollno: 1003
Name: mani
Marks: 98

ALL